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$$\text{chance} = \frac{P}{1-P} \quad \begin{array}{l} \text{evento} \\ \text{NÃO evento} \end{array}$$

$$P_{\text{AMANDA}} = 0,80 \Rightarrow \text{chance} = \frac{0,80}{0,20} = \frac{4}{1} = 4$$

$$P_{\text{LUIZ}} = 0,25 \Rightarrow \text{chance} = \frac{0,25}{0,75} = \frac{1}{3}$$

$$\text{OLS: } \hat{Y}_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_K X_{Ki}$$

Logística Binária (Hosmer & Lemeshow: Communications in Statistics (1989))

$$\ln\left(\frac{p_i}{1-p_i}\right) = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_K X_{Ki}$$

z : logito

$$\ln\left(\frac{p}{1-p}\right) = z$$

$$\frac{p}{1-p} = e^z$$

$$p = e^z - p \cdot e^z$$

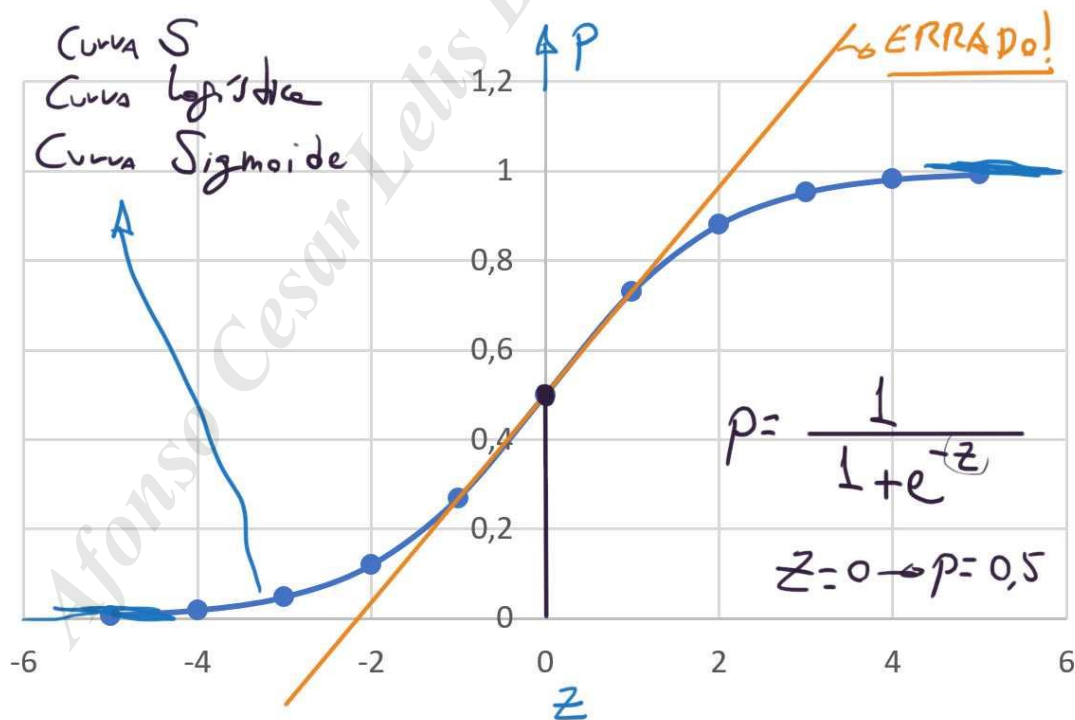
$$p(1 + e^z) = e^z$$

$$\hat{p} = \frac{e^z}{1 + e^z} \quad \parallel \quad \hat{p} = \frac{1}{1 + e^{-z}}$$

$$-\infty \leq z \leq +\infty$$

$$\hat{p}_i = \frac{1}{1 + e^{-(\alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_K X_{Ki})}}$$

$$\underbrace{\frac{1}{1+e^{-(z)}}}_{P_{\text{evento}}} + \underbrace{\frac{1}{1+e^{(z)}}}_{P_{\text{NÃO evento}}} = 1$$



sim (evento)
NÃO (evento)

$$Y = 1(\text{evento})$$

$$Y = 0(\sim \text{evento})$$

dist

sem

evento: cat. Alter.
NÃO evento: cat. Ref.

Atrasado
quali

$$P_{\text{atrasado}_i} = \frac{1}{1 + e^{-(\alpha + \beta_1 \text{dist}_i + \beta_2 \text{sem}_i)}}$$

$$\text{atrasado} \sim \text{Bernoulli}(\text{prob}_{\text{atrasado}=1} = \hat{P})$$

$$\log \left[\frac{\hat{P}}{1 - \hat{P}} \right] = -26.17 + 0.19(\text{dist}) + 2.36(\text{sem})$$

$$\hat{P}_i = \frac{1}{1 + e^{-(-26,17 + 0,19 \text{dist}_i + 2,36 \text{sem}_i)}}$$

Bernoulli

$$p(Y_i) = p_i^{Y_i} \cdot (1 - p_i)^{1 - Y_i}$$

$$p(Y=1) = p^1 \cdot (1-p)^{1-1} = \underline{p}$$

$$p(Y=0) = p^0 \cdot (1-p)^{1-0} = \underline{1-p}$$

$$\chi^2 \sim F$$

$$\uparrow LL \quad \downarrow \begin{matrix} AIC \\ BIC \end{matrix}$$

$$\underline{\chi^2 = -2 \cdot (LL_0 - LL_{\text{modelo}})}$$

AIC (Akaike Info Criterion):

$$AIC = -2 \underline{LL_m} + 2(\underline{k} + 1)$$

BIC (Bayesian Info Criterion):

$$BIC = -2 \underline{LL_m} + (\underline{k} + 1) \cdot \ln(n)$$

melhor,
melhor

pseudo R^2 McFadden (Nobel Economia 2000)
(cellas discretas)

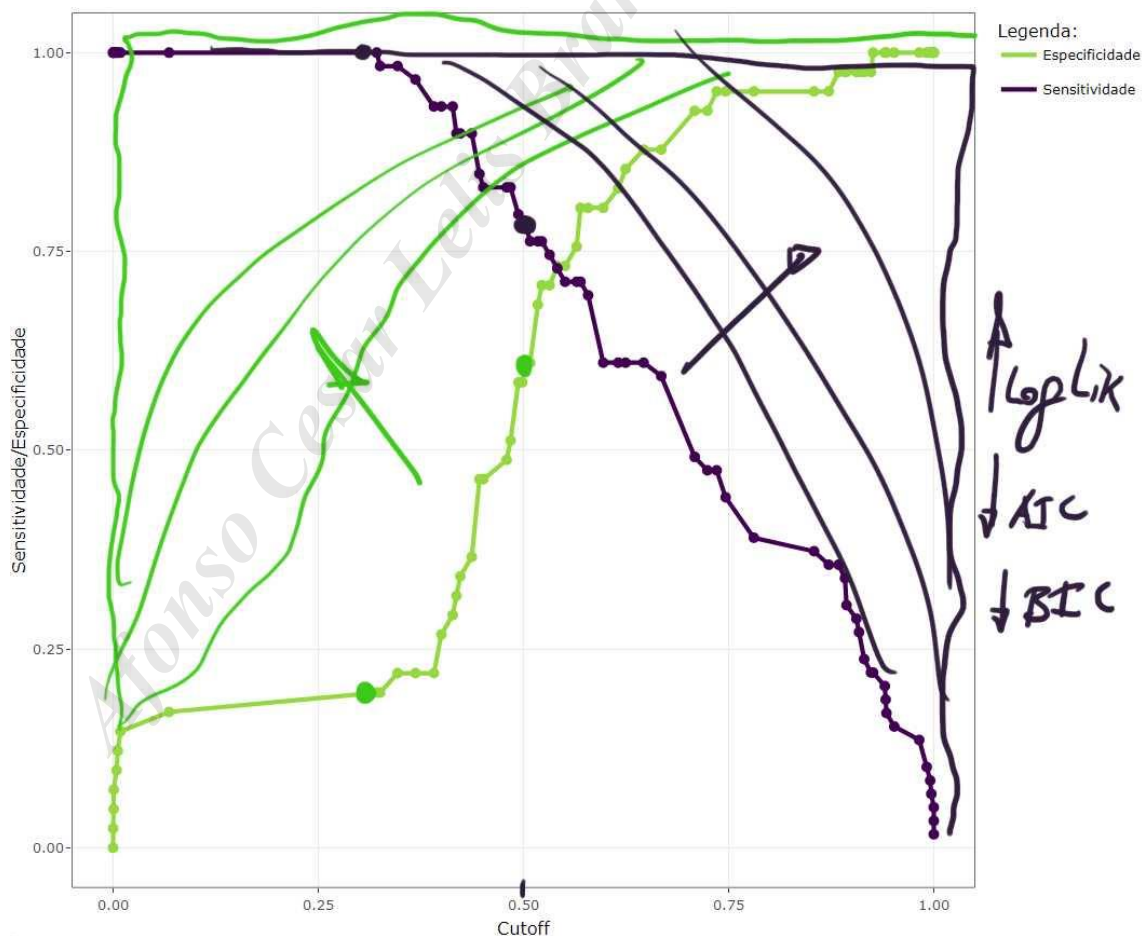
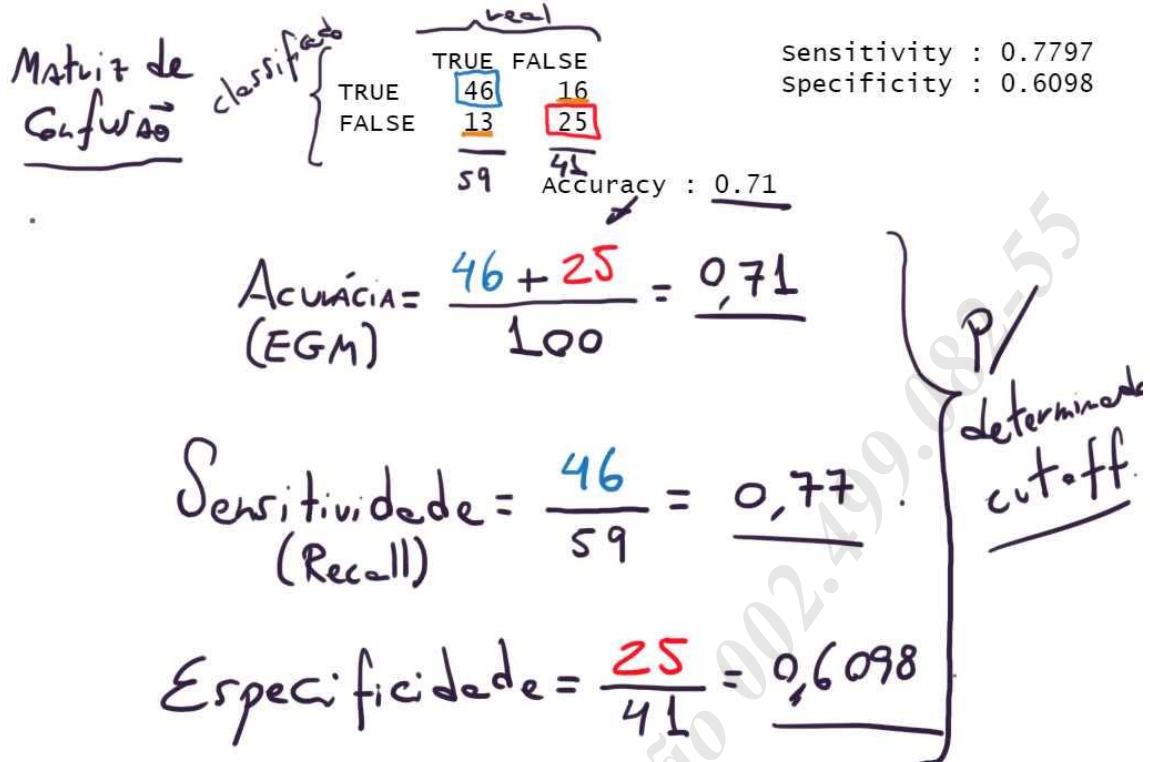
$$P_{MF}^{R^2} = \frac{\overbrace{-2 \cdot (U_0 - U_{\text{modelo}})}^{x^2}}{-2U_0}$$

$$P_{C-U}^{R^2} \text{ (Nagelkerke)} = \frac{1 - \left(\frac{e^{U_0}}{e^{U_L}} \right)^{\frac{2}{n}}}{1 - (e^{U_0})^{\frac{2}{N}}}$$

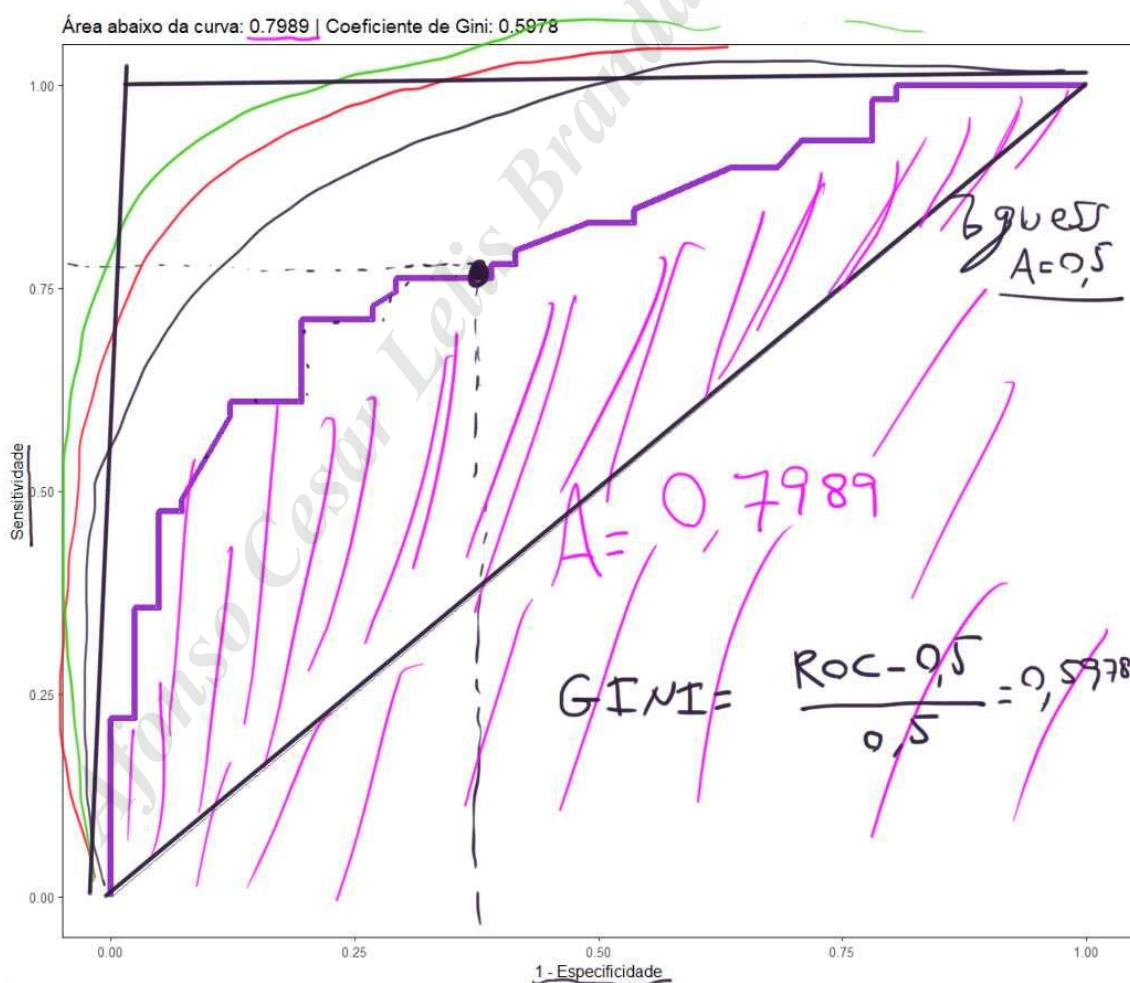
cutoff:

Se $phat \geq \text{cutoff} \Rightarrow$ EVENTO

Se $phat < \text{cutoff} \Rightarrow$ NÃO evento



ROC: Receiver Operating
Characteristic



$$p = \frac{1}{1 + e^{-(z)}}$$

cutoff

matriz de confusão

Acurácia (EGM) }
Sensitividade } p/ aquele
Especificidade } cutoff

Log Lik ↑
AIC ↓
BIC ↓
ROC ↑
GINI ↑

} independentes do cutoff